

MTH102

Linear Algebra

2025–2026–I SEMESTER

PROBLEM SETS

09 October 2025

Chapter 1. Problem set 1

Exercise 1.1. Determine the elements of the following sets.

- $\{x \in \mathbb{N} : x^2 - 1 = 0\}$.
- $\{x \in \mathbb{Z} : x^2 - 1 = 0\}$.

Exercise 1.2. What does it mean to say that A is not a subset of B ?

Exercise 1.3. Show that if A is a subset of B and B is a subset of C , then A is a subset of C .

Exercise 1.4. Show that no element n of $\mathbb{N}$ satisfies $n^4 - 5n^2 + 6 = 0$.

Exercise 1.5. What does it mean to say that two sets A, B are not equal?

Exercise 1.6. Consider the sets

$$\{1, 2, 3, 4\}, \{3, 4, 5, 6\}.$$

Determine the elements common to these sets.

Exercise 1.7. Determine the union of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

Exercise 1.8. If A, B are sets, show that

$$A \subseteq A \cup B, B \subseteq A \cup B.$$

Exercise 1.9. If A, B are subsets of a set C , then $A \cup B$ is a subset of C .

Exercise 1.10. If A is a set, show that

$$A \cup A = A.$$

Exercise 1.11 (Commutative property). If A, B are sets, show that

$$A \cup B = B \cup A.$$

Exercise 1.12 (Associative property). If A, B, C are sets, show that

$$A \cup B \cup C = (A \cup B) \cup C = A \cup (B \cup C).$$

Exercise 1.13. Determine the intersection of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

Exercise 1.14. If A, B are sets, show that

$$A \cap B \subseteq A, A \cap B \subseteq B.$$

Exercise 1.15. If A, B, C are sets satisfying

$$C \subseteq A, C \subseteq B,$$

then show that

$$C \subseteq A \cap B$$

holds.

Exercise 1.16. Identify the integers among

$$1, 2, 3, 4, \dots, 20$$

which are divisible by at least one of 2 and 5.

Exercise 1.17 (Inclusion-exclusion principle). Show that for finite subsets A, B, C of a set X ,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.$$

Exercise 1.18. Determine the number of integers among

$$1, 2, 3, 4, \dots, 100,$$

which are divisible by at least one of 6, 10, 15.

Exercise 1.19. If A is a set, show that

$$A \cap A = A.$$

Exercise 1.20 (Commutative property). If A, B are sets, show that

$$A \cap B = B \cap A.$$

Exercise 1.21 (Associative property). If A, B, C are sets, show that

$$A \cap B \cap C = (A \cap B) \cap C = A \cap (B \cap C).$$

Exercise 1.22. Show that

$$A^c = \{x \in X : x \notin A\},$$

where X denotes the underlying universal set.

Exercise 1.23. Determine the complement of $\{1, 2, 3, 4\}$ in $\{1, 3, 5, 7\}$, and the complement of $\{1, 3, 5, 7\}$ in $\{1, 2, 3, 4\}$.

Exercise 1.24. If A, B are subsets of a set X , show that

$$A^c \cap B = B \setminus A.$$

Exercise 1.25. If A is a subset of a set X , then show that

$$A \cup A^c = X, A \cap A^c = \emptyset, (A^c)^c = A.$$

Exercise 1.26. If A, B are subsets of a set X , then show that

$$(A \cup B)^c = A^c \cap B^c, (A \cap B)^c = A^c \cup B^c.$$

Exercise 1.27. Let A, B be subsets of a set X . Show that the following statements are equivalent.

$$\begin{aligned} A &\subseteq B, \\ A \cap B &= A, \\ A \cup B &= B, \\ B^c &\subseteq A^c. \end{aligned}$$

Exercise 1.28. Let A, B be subsets of a set X . Show that the sets

$$A \setminus B, B \setminus A$$

are disjoint.

Exercise 1.29. Determine the symmetric difference of $\{1, 2, 3, 4\}$ and $\{1, 3, 5, 7\}$.

Exercise 1.30. If A, B are sets, then show that

$$\begin{aligned} A \Delta B &= (A \setminus B) \cup (B \setminus A), \\ A \Delta B &= B \Delta A, \\ A \cup B &= (A \Delta B) \cup (A \cap B), \\ (A \Delta B) \cap (A \cap B) &= \emptyset. \end{aligned}$$

Exercise 1.31. Write down the power set of each of the following sets:

$$\{1, 2\}, \{1, 2, 3\}, \{1, \{2, 3\}\}, \{1, 2, 3, 4\}.$$

Exercise 1.32. If A, B are finite sets, then show that

$$|A \times B| = |A| \times |B|,$$

where for a set X , its number of elements is denoted by $|X|$.

Exercise 1.33. Determine the cartesian product of $[1, 2]$ and $[3, 4] \cup [5, 6]$.

Exercise 1.34. Identify the set

$$\left\{x \in \mathbb{R} \setminus \{0\} : x + \frac{1}{x} \geq 2\right\}.$$

Exercise 1.35. Let A, B be subsets of $\mathbb{R}$ defined by

$$\begin{aligned} A &= \{x \in \mathbb{R} : x^2 \geq 0\}, \\ B &= \{x \in \mathbb{R} : x^3 \geq 0\}. \end{aligned}$$

Determine the sets $A \cup B, A \cap B, A \setminus B, B \setminus A$.

Exercise 1.36. For a positive integer k , let A_k denote the set of integral multiples of k , that is,

$$A_k := \{r \in \mathbb{Z} : r = k\ell \text{ for some } \ell \in \mathbb{Z}\}.$$

Let m, n be positive integers. For each of the following statements, determine the equivalent conditions on the integers m, n .

1. $A_m \subseteq A_n$
2. $A_m \subsetneq A_n$
3. $A_m \not\subseteq A_n$
4. $A_m = A_n$
5. $A_m \cap A_n = \emptyset$
6. $A_m \setminus A_n \neq \emptyset$
7. $A_m \setminus A_n \neq \emptyset$ or $A_n \setminus A_m \neq \emptyset$
8. $A_m \setminus A_n \neq \emptyset$ and $A_n \setminus A_m \neq \emptyset$

Exercise 1.37. Show that $\mathbb{R}$ is a subset of $\mathbb{C}$.

Exercise 1.38. If z, w are complex numbers, then show that

1. $\overline{z + w} = \overline{z} + \overline{w}$,
2. $\overline{z \cdot w} = \overline{z} \cdot \overline{w}$,
3. $|z| \geq 0$, and $|z| = 0$ if and only if $z = 0$,
4. $|z \cdot w| = |z| |w|$.

Exercise 1.39. If z is a complex number, then show that

$$z \cdot \overline{z} = |z|^2.$$

Exercise 1.40. For any real number θ , show that

$$|\cos \theta + i \sin \theta| = 1.$$

Chapter 2. Induction principles

Exercise 2.1. If A is a nonempty subset of $\mathbb{N}$, and a_1, a_2 are least elements of A , then show that $a_1 = a_2$.

Exercise 2.2. Show that

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

for all $n \in \mathbb{N}$.

Exercise 2.3. Show that

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

for all $n \in \mathbb{N}$.

Exercise 2.4. Show that

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \cdots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$$

for all $n \in \mathbb{N}$.

Exercise 2.5. Show that

$$3 + 11 + \cdots + (8n-5) = 4n^2 - n$$

for all $n \in \mathbb{N}$.

Exercise 2.6. Show that

$$1^2 + 3^2 + \cdots + (2n-1)^2 = \frac{4n^3 - n}{3}$$

for all $n \in \mathbb{N}$.

Exercise 2.7. Show that

$$1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{n+1}n^2 = (-1)^{n+1} \frac{n(n+1)}{2}$$

for all $n \in \mathbb{N}$.

Exercise 2.8. Show that

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6},$$

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

hold for any positive integer n .

Exercise 2.9. Show that

$$\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2n-1} - \frac{1}{2n}$$

for all $n \in \mathbb{N}$.

Exercise 2.10. Show that $n^3 + 5n$ is divisible by 6 for all $n \in \mathbb{N}$.

Exercise 2.11. Show that $5^{2n} - 1$ is divisible by 8 for all $n \in \mathbb{N}$.

Exercise 2.12. Show that $5^n - 4n - 1$ is divisible by 16 for all $n \in \mathbb{N}$.

Exercise 2.13. Show that $6^n - 5n - 1$ is divisible by 25 for all $n \in \mathbb{N}$.

Exercise 2.14. Show that $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9 for all $n \in \mathbb{N}$.

Exercise 2.15. Show that

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

for all $n \in \mathbb{N}$ satisfying $n > 1$.

Exercise 2.16. Show that $3^n \geq n^2$ for all $n \in \mathbb{N}$.

Exercise 2.17. Show that $n! > 2^n$ for all $n \in \mathbb{N}$ satisfying $n \geq 4$.

Here are a few exercises where strong induction is useful.

Exercise 2.18. Show that every positive integer greater than 1 is a prime or is a product of prime numbers.

Exercise 2.19. Let the integers $x_1, x_2, \dots$ be defined by

$$x_1 := 1,$$

$$x_2 := 2,$$

$$x_{n+2} := \frac{1}{2}(x_n + x_{n+1})$$

for all $n \in \mathbb{N}$. Show that $1 \leq x_n \leq 2$ for all $n \in \mathbb{N}$.

Exercise 2.20. The Fibonacci sequence $F_0, F_1, F_2, \dots$ is defined by

$$F_0 := 0,$$

$$F_1 := 1,$$

$$F_2 := 1,$$

$$F_{n+2} := F_n + F_{n+1}$$

for all $n \in \mathbb{N}$. Show that

$$F_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right)$$

for all $n \in \mathbb{N} \cup \{0\}$.

Exercise 2.21. Show that

$$F_0^2 + F_1^2 + F_2^2 + \cdots + F_n^2 = F_n F_{n+1}$$

for all $n \in \mathbb{N} \cup \{0\}$.

Chapter 3. Matrices

Exercise 3.1. If A, B, C are matrices of the same size, show that

$$A + (B + C) = (A + B) + C.$$

Exercise 3.2. If A, B are matrices of the same size, show that

$$A + B = B + A.$$

Exercise 3.3. If A is an $m \times n$ matrix, show that

$$A + 0_{m \times n} = 0_{m \times n} + A = A$$

holds, where $0_{m \times n}$ denotes the $m \times n$ zero matrix, that is, the $m \times n$ matrix whose entries are all zero.

Exercise 3.4. Show that if A is a matrix and c, d are real numbers, then

$$(c + d)A = cA + dA$$

and

$$c(dA) = (cd)A.$$

Exercise 3.5. If A is a matrix, what is $(A^T)^T$?

Exercise 3.6. Show that if A, B are matrices of the same size, then

$$(A + B)^T = A^T + B^T.$$

Exercise 3.7. Show that if A denotes a matrix, then

$$(A^T)^T = A.$$

Exercise 3.8. Show that if A, B, C are matrices such that the products $A(BC)$ and $(AB)C$ are defined, then

$$A(BC) = (AB)C.$$

Exercise 3.9. Provide examples to show that in general, matrix multiplication is not commutative, that is, $AB \neq BA$ for some matrices A, B .

Exercise 3.10 ().** Show that for any $n \in \mathbb{N}$ with $n \geq 2$, there are $n \times n$ matrices A, B such that

$$AB \neq BA.$$

? Question

Can induction be used for the above exercise?

Exercise 3.11. If A is an $n \times n$ matrix, show that

$$AI_n = I_n A = A$$

holds, where I_n denotes the $n \times n$ diagonal matrix, with all diagonal entries equal to 1, that is,

$$I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}.$$

Exercise 3.12. If A is an $m \times n$ matrix, show that

$$I_m A = A, A I_n = A$$

where I_m (resp. I_n) denote the $m \times m$ (resp. $n \times n$) identity matrix.

Exercise 3.13. If A is an $m \times n$ matrix with entries in $\mathbb{R}$ and c is a real number, then

$$cA = (cI_n)A = A(cI_m),$$

where I_n (resp. I_m) denotes the $n \times n$ (resp. $m \times m$) identity matrix.

Exercise 3.14. Show that if A, B are matrices, then

$$(AB)^T = B^T A^T.$$

Exercise 3.15. Show that if

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix},$$

then

$$A^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}$$

for all $n \in \mathbb{N}$ with $n \geq 1$, where F_n denotes the n -th Fibonacci number.

Exercise 3.16. Compute

$$\begin{pmatrix} 0 & \dots & 0 & 1 \\ 0 & \dots & 1 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 1 & \dots & 0 & 0 \end{pmatrix}^2.$$

Exercise 3.17. Show that if A, B, C are matrices such that the sum $B + C$, and the products $AB, AC, A(B + C)$ are defined, then

$$A(B + C) = AB + AC.$$

Exercise 3.18. Show that if A, B, C are matrices such that the sum $B + C$, and the products $AB, AC, A(B + C)$ are defined, then

$$(A + B)C = AC + BC.$$

Exercise 3.19. If A is an $m \times n$ matrix, then show that for all $i = 1, 2, \dots, n$, the product Ae_i is equal to the i -th column of A , where e_i denotes the i -th standard basis vector of $\mathbb{R}^n$, that is,

$$e_i = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$

with 1 in the i -th position, and 0's elsewhere.

Exercise 3.20. Determine whether the identity matrix I_n is invertible, and if it is, find its inverse.

Exercise 3.21. If A is an invertible square matrix, then show that A^{-1} is also invertible, and that

$$(A^{-1})^{-1} = A.$$

Exercise 3.22. Let A, B be invertible $n \times n$ matrices. Show that AB is invertible, and that

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Compare the above exercise with [Exercise 3.14](#).

Exercise 3.23. If A, B are 2×2 matrices, then show that

$$\det(AB) = \det(A) \det(B).$$

Exercise 3.24. Solve the system of equations

$$\begin{aligned} 5x + 2y &= 11, \\ 3x + 4y &= 8. \end{aligned}$$

Exercise 3.25. Solve the system of equations

$$\begin{aligned} 12x - 25y &= -47, \\ -7x + 30y &= 51. \end{aligned}$$

Exercise 3.26. Solve the system of equations

$$\begin{aligned} 2x + 3y + z &= 1, \\ 4x + y + 2z &= 2, \\ 3x + 2y + 3z &= 3. \end{aligned}$$

Exercise 3.27. Let A, B be 2×2 matrices. Show that if AB is invertible, then both A and B are invertible.

Compare the above with [Exercise 3.63](#), [Exercise 3.64](#), [Exercise 3.75](#).

Exercise 3.28. Let A be an $n \times n$ matrix. If P is an invertible $n \times n$ matrix, then show that

$$(PAP^{-1})^k = PA^kP^{-1}$$

holds for any positive integer k .

Exercise 3.29. Let A be a 2×2 matrix. Let V be the set of solutions to the system of equations

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

that is,

$$V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}.$$

Show that for any $u, v \in V$ and $c \in \mathbb{R}$, the element $u + v$ of $\mathbb{R}^2$ lies in V , and the element cu also lies in V .

Exercise 3.30. Let A be a 2×2 matrix. Let V be the set of vectors which can be expressed as a linear combination of the columns of A , that is¹,

$$V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : \begin{pmatrix} x \\ y \end{pmatrix} = A \begin{pmatrix} s \\ t \end{pmatrix} \text{ for some } s, t \in \mathbb{R} \right\}.$$

Show that for any $u, v \in V$ and $c \in \mathbb{R}$, the element $u + v$ of $\mathbb{R}^2$ lies in V , and the element cu also lies in V .

Compare the following exercise with [Exercise 3.57](#).

Exercise 3.31. Let A be an $m \times n$ matrix. Let V be the set of solutions to the system of equations

$$Au = 0,$$

that is,

$$V = \{v \in \mathbb{R}^n : Av = 0\}.$$

Show that for any $u, v \in V$ and $c, d \in \mathbb{R}$, the element $cu + dv$ of $\mathbb{R}^m$ lies in V .

Compare the following exercise with [Exercise 3.58](#).

Exercise 3.32. Let A be an $m \times n$ matrix. Let V be the set of vectors which can be expressed as a linear combination of the columns of A , that is,

$$V = \{v \in \mathbb{R}^m : v = Au \text{ for some } u \in \mathbb{R}^n\}.$$

Show that for any $u, v \in V$ and $c, d \in \mathbb{R}$, the element $cu + dv$ of $\mathbb{R}^m$ lies in V .

Exercise 3.33. Let A be a square matrix. Show that if A is symmetric and skew-symmetric, then A is the zero matrix.

Exercise 3.34. Show that for any $n \geq 2$, there are infinitely many $n \times n$ orthogonal matrices.

Exercise 3.35. Let A be an orthogonal matrix. Show that A is invertible and that

$$A^{-1} = A^T.$$

Exercise 3.36. Let A, B be orthogonal matrices of the same size. Show that the matrix product AB is also an orthogonal matrix.

Exercise 3.37. Let A, B be square matrices with complex entries. Show that

$$\overline{AB} = \overline{A} \overline{B}.$$

¹Observe that

$$A \begin{pmatrix} s \\ t \end{pmatrix} = sC_1 + tC_2,$$

where C_1, C_2 denote the first and second columns of A respectively.

Exercise 3.38. Let A be a square matrix with complex entries. Show that

$$\overline{A^T} = (\overline{A})^T.$$

Exercise 3.39. Let A, B be square matrices of the same size with complex entries. Show that

$$(A + B)^* = A^* + B^*.$$

Exercise 3.40. Let A be a matrix with complex entries. Show that

$$(A^*)^* = A.$$

Exercise 3.41. Let A, B be matrices with complex entries. Show that

$$(AB)^* = B^* A^*.$$

Exercise 3.42. Let A be an $n \times n$ hermitian matrix. Show that all the diagonal entries of A are real numbers. More generally, show that for any $1 \leq i, j \leq n$, the (i, j) -entry and the (j, i) -entry of A are complex conjugates of each other.

Exercise 3.43. Let A, B be hermitian matrices of the same size. Show that the matrix $A + B$ is also a hermitian matrix. Is the matrix AB also a hermitian matrix?

Exercise 3.44. State and prove an analogue of [Exercise 3.42](#) for skew-hermitian matrices.

Exercise 3.45. Let A, B be skew-hermitian matrices of the same size. Show that the matrix $A + B$ is also a skew-hermitian matrix. Is the matrix AB also a skew-hermitian matrix?

Exercise 3.46. Let A be a square matrix with complex entries. Show that if A is hermitian and skew-hermitian, then A is the zero matrix.

Exercise 3.47. Let A be a unitary matrix. Show that A is invertible and that

$$A^{-1} = A^*.$$

Exercise 3.48. Let A, B be unitary matrices of the same size. Show that the matrix product AB is also a unitary matrix.

Exercise 3.49. Show that any orthogonal matrix is a unitary matrix. Is the converse true?

Exercise 3.50. Determine whether the following matrices are unitary, hermitian, both, or neither.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Exercise 3.51. Let z, w be complex numbers such that $|z|^2 + |w|^2 = 1$. Show that

$$\begin{pmatrix} z & -\overline{w} \\ w & \overline{z} \end{pmatrix}$$

is a unitary matrix.

Exercise 3.52. Let A be a diagonal matrix with complex entries. Show that A is a unitary matrix if and only if all the diagonal entries of A have absolute value equal to 1.

Exercise 3.53. Show that any unitary matrix is a normal matrix. Is the converse true?

Exercise 3.54. Show that any hermitian matrix is a normal matrix. Is the converse true?

Exercise 3.55. Show that any skew-hermitian matrix is a normal matrix. Is the converse true?

Exercise 3.56. Let e_{ij}, e_{kl} be matrix units of the same size. Show that $e_{ij}e_{kl}$ is equal to the matrix unit e_{il} if $j = k$, and is equal to the zero matrix if $j \neq k$, that is,

$$e_{ij}e_{kl} = \begin{cases} e_{il} & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases}$$

Denote the set of $m \times n$ matrices with entries from $\mathbb{R}$ by $M_{m,n}(\mathbb{R})$.

Compare the following exercise with [Exercise 3.31](#).

Exercise 3.57. Let A be an $m \times n$ matrix. Let V be the set of solutions to the system of equations

$$uA = 0,$$

that is,

$$V = \{v \in M_{1,m}(\mathbb{R}) : vA = 0\}.$$

Show that for any $u, v \in V$ and $c, d \in \mathbb{R}$, the element $cu + dv$ of $\mathbb{R}^m$ lies in V .

Compare the following exercise with [Exercise 3.32](#).

Exercise 3.58. Let A be an $m \times n$ matrix. Let V be the set of row vectors which can be expressed as a linear combination of the rows of A , that is,

$$V = \{v \in M_{1,n}(\mathbb{R}) : v = u^T A \text{ for some } u \in \mathbb{R}^m\}.$$

Show that for any $u, v \in V$ and $c, d \in \mathbb{R}$, the element $cu + dv$ of $M_{1,n}(\mathbb{R})$ lies in V .

Exercise 3.59. Solve the system of linear equations in five variables x_1, x_2, x_3, x_4, x_5 :

$$\begin{aligned} x_1 + 2x_2 - x_3 + 4x_4 + x_5 &= 7, \\ 2x_1 + 3x_2 + x_3 + 5x_4 + 2x_5 &= 14, \\ -x_1 + 4x_2 - 2x_3 - 3x_4 + x_5 &= -10. \end{aligned} \tag{1}$$

Exercise 3.60. Solve the system of equations in the variables x_1, x_2, x_3 :

$$\begin{aligned} x_1 + 2x_2 - x_3 &= 1, \\ 2x_1 + 3x_2 + x_3 &= 2, \\ -x_1 + 4x_2 - 2x_3 &= -3, \\ 3x_1 - x_2 + 4x_3 &= 4, \\ 5x_1 + 2x_2 + 3x_3 &= 5. \end{aligned}$$

Exercise 3.61. Solve the system of equations in the variables x_1, x_2, x_3, x_4 :

$$\begin{aligned}
x_1 + 2x_2 - x_3 + 4x_4 &= 5, \\
2x_1 + 3x_2 + x_3 + 5x_4 &= 8, \\
-x_1 + 4x_2 - 2x_3 - 3x_4 &= -4, \\
3x_1 - x_2 + 4x_3 + 2x_4 &= 7.
\end{aligned}$$

Exercise 3.62. Solve the system of equations in the variables x_1, x_2, x_3 :

$$\begin{aligned}
x_1 + 2x_2 - 5x_3 &= 20, \\
2x_1 + 5x_2 - 7x_3 &= 33, \\
-x_1 - 2x_2 + 4x_3 &= -17.
\end{aligned}$$

Exercise 3.63. Let A, B be square matrices of the same size. Suppose that A is invertible and $AB = I$ holds. Show that B is invertible and $B = A^{-1}$.

Exercise 3.64. Let A, B be square matrices of the same size. Suppose that A is invertible and $BA = I$ holds. Show that B is invertible and $B = A^{-1}$.

Compare the above two exercises with [Exercise 3.27](#), [Exercise 3.75](#).

Exercise 3.65. Let A, B be matrices, suppose that the number of columns of A is equal to the number of rows of B . Suppose AB has at least two columns. Let C be the matrix obtained from AB by removing the last column of AB . Show that there exists a matrix B' such that $C = AB'$.

Exercise 3.66. Let A, B be matrices, suppose that the number of columns of A is equal to the number of rows of B . Suppose AB has at least two rows. Let C be the matrix obtained from AB by removing the last row of AB . Show that there exists a matrix A' such that $C = A'B$.

Exercise 3.67. Use elementary row operations to reduce the matrix

$$A = \begin{pmatrix} 1 & 2 & -5 \\ 2 & 5 & -7 \\ -1 & -2 & 4 \end{pmatrix}$$

in its row echelon form. Use the row echelon form to determine whether A is invertible. If A is invertible, use elementary matrices corresponding to the performed elementary row operations to find A^{-1} .

Exercise 3.68. Solve the system of equations in the variables x_1, x_2 :

$$\begin{aligned}
2x_1 - 3x_2 &= 7, \\
-4x_1 + 5x_2 &= -11.
\end{aligned}$$

Exercise 3.69. Transform the matrix

$$\begin{pmatrix} 0 & 2 & -3 & 0 & 0 \\ 1 & -1 & 4 & 0 & 5 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 3 & -5 & 0 & 1 \\ 0 & 3 & -5 & -1 & 2 \end{pmatrix}$$

into a matrix in row echelon form using elementary row operations.

Exercise 3.70. Solve the system of equations in the variables x_1, x_2, x_3, x_4 :

$$\begin{aligned} 2x_2 - 3x_3 &= 0, \\ x_1 - x_2 + 4x_3 &= 5, \\ x_4 &= -2, \\ 3x_2 - 5x_3 &= 1, \\ 3x_2 - 5x_3 - x_4 &= 2. \end{aligned}$$

Exercise 3.71. Find the rank of the matrix

$$\begin{pmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 1 & -\frac{2}{15} \\ 0 & 0 & 1 & -\frac{1}{14} \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Exercise 3.72. If A is an $m \times n$ matrix, then show that

$$\text{rk}(A) \leq \min\{m, n\}.$$

Exercise 3.73. If A is an $n \times n$ matrix, then show that the following statements are equivalent.

1. The matrix A is invertible.
2. The rank of A is n .

Exercise 3.74. If A is an invertible square matrix, then show that A^{-1} is also invertible, and that

$$(A^{-1})^{-1} = A.$$

Exercise 3.75. If A, B are square matrices of the same size satisfying $BA = I$, then show that the matrices A, B are invertible, and are the inverses of each other.

Compare the above with [Exercise 3.63](#), [Exercise 3.64](#), [Exercise 3.27](#).

Exercise 3.76. Solve the system of equations in the variables x, y, z :

$$\begin{aligned} x + 2y + 3z &= 1, \\ 4x + 5y + 6z &= 2, \\ 7x + 8y + 10z &= 3. \end{aligned}$$

Exercise 3.77. Show that the determinant of the identity matrix I_3 is 1, and that

$$\text{adj}(I_3) = I_3.$$

Exercise 3.78. Let

$$A = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix},$$

where a, b, c are scalars. Show that $\det(A) = abc$, and that

$$\operatorname{adj}(A) = \begin{pmatrix} bc & 0 & 0 \\ 0 & ca & 0 \\ 0 & 0 & ab \end{pmatrix}.$$

Exercise 3.79. Let

$$A = \begin{pmatrix} 1 & -5 & 6 \\ -4 & 8 & -9 \\ 7 & 2 & 10 \end{pmatrix}.$$

Find $\det(A)$ and $\operatorname{adj}(A)$.

Exercise 3.80. Let

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 3 & 2 & 1 \\ 4 & 1 & 2 \end{pmatrix}.$$

Find $\det(A)$ and $\operatorname{adj}(A)$.

Exercise 3.81. Let

$$A = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$$

where a, b are scalars. Show that $\det(A) = ab$, and that

$$\operatorname{adj}(A) = \begin{pmatrix} b & 0 \\ 0 & a \end{pmatrix}.$$

Exercise 3.82. Compute the determinant of the matrix

$$A = \begin{pmatrix} 1 & -5 & 6 \\ 2 & 0 & 9 \\ -1 & 2 & -1 \end{pmatrix}$$

to determine whether it is invertible. If A is invertible, then find its inverse using its adjoint.

Exercise 3.83. Use Gaussian elimination to determine the row echelon form of the matrix

$$A = \begin{pmatrix} 1 & -5 & 6 \\ 2 & 0 & 9 \\ -1 & 2 & -1 \end{pmatrix}.$$

Use the row echelon form to determine whether A is invertible. If A is invertible, then find its inverse using elementary matrices, corresponding to the elementary row operations used to transform A into its row echelon form.

Exercise 3.84. Solve the following system of equations in the variables x, y, z .

$$\begin{aligned} x + 2y + 3z &= 1, \\ 4x - 5y + 6z &= 2, \\ 7x - y + 10z &= 3. \end{aligned}$$

Chapter 4. The space $\mathbb{R}^n$

Exercise 4.1. Let S be a subset of $\mathbb{R}^n$, containing precisely one element v . Show that S is linearly independent if and only if $v \neq 0$.

Exercise 4.2. In each of the following, determine whether the given vectors are linearly independent or dependent. Justify your answers.

1. $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$, and $\begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$ in $\mathbb{R}^3$.
2. $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, and $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ in $\mathbb{R}^3$.
3. $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ in $\mathbb{R}^2$.
4. $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ in $\mathbb{R}^2$.
5. $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$, and $\begin{pmatrix} 7 \\ 8 \\ 10 \end{pmatrix}$ in $\mathbb{R}^3$.

Exercise 4.3. Let $v_1 = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$, $v_2 = \begin{pmatrix} 4 \\ 0 \\ -1 \end{pmatrix}$, $v_3 = \begin{pmatrix} -1 \\ 5 \\ 2 \end{pmatrix}$. Determine whether the vectors v_1, v_2, v_3 are linearly independent or dependent. Justify your answer.

Exercise 4.4. Show that the standard basis vectors

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, e_n := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

of $\mathbb{R}^n$ are linearly independent.

Exercise 4.5. Show that any finite subset of $\mathbb{R}^n$ containing the zero vector is linearly dependent.

Exercise 4.6. Show that any subset of a linearly independent set of vectors is also linearly independent.

Exercise 4.7. Let $v_1, v_2, \dots, v_k$ be elements of $\mathbb{R}^n$. If $k > n$, then the vectors $v_1, v_2, \dots, v_k$ are linearly dependent, that is, at least one of the vectors can be expressed as a linear combination of the others.

Exercise 4.8. Let V denote the set of all vectors of $\mathbb{R}^n$ whose first component is equal to zero, that is,

$$V = \left\{ \begin{pmatrix} 0 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix} : x_i \in \mathbb{R} \text{ for all } i = 2, 3, \dots, n \right\}.$$

Show that V is a linear subspace of $\mathbb{R}^n$.

Exercise 4.9. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}$ is a linear subspace of $\mathbb{R}^3$.

Exercise 4.10. Let (a, b, c) be a nonzero element of $\mathbb{R}^3$. Show that the set

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : ax + by + cz = 0 \right\}$$

is a linear subspace of $\mathbb{R}^3$.

Exercise 4.11. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x \geq 0, y \geq 0 \right\}$ is not a linear subspace of $\mathbb{R}^2$.

Exercise 4.12. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x = 1 \right\}$ is not a linear subspace of $\mathbb{R}^2$.

Exercise 4.13. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x = y \right\}$ is a linear subspace of $\mathbb{R}^2$.

Exercise 4.14. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : z = 0 \right\}$ is a linear subspace of $\mathbb{R}^3$.

Exercise 4.15. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 1 \right\}$ is not a linear subspace of $\mathbb{R}^3$.

Exercise 4.16. Show that $\mathbb{R}^n$ is a linear subspace of itself.

Exercise 4.17. Show that the set containing only the zero vector, that is, the set $\{0\}$, is a linear subspace of $\mathbb{R}^n$.

Exercise 4.18. Let A be an $m \times n$ matrix. Let V be the set of solutions to the system of equations

$$Au = 0,$$

that is,

$$V = \{v \in \mathbb{R}^n : Av = 0\}.$$

Show that V is a linear subspace of $\mathbb{R}^n$.

Exercise 4.19. Let A be an $m \times n$ matrix. Let V be the set of vectors which can be expressed as a linear combination of the columns of A , that is,

$$V = \{v \in \mathbb{R}^m : v = Au \text{ for some } u \in \mathbb{R}^n\}.$$

Show that V is a linear subspace of $\mathbb{R}^m$.

Exercise 4.20. Let A be an $m \times n$ matrix with real entries. Show that for any invertible $m \times m$ matrix E , the null space $\text{null}(A)$ is equal to the null space $\text{null}(EA)$.

Exercise 4.21. Let m, n be positive integers with $m \geq 2$. Provide an example of an $m \times n$ matrix A with real entries and an invertible $m \times m$ matrix E such that the column space $\text{col}(A)$ of A is not equal to the column space $\text{col}(EA)$ of EA .

Exercise 4.22. Let U, V be the subsets of $\mathbb{R}^3$ defined as

$$U = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + 2y = 0 \right\},$$

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}.$$

Show that both U and V are linear subspaces of $\mathbb{R}^3$. Determine the intersection $U \cap V$ of U and V . Is the intersection $U \cap V$ a linear subspace of $\mathbb{R}^3$?

Exercise 4.23. Show that the intersection of two linear subspaces of $\mathbb{R}^n$ is also a linear subspace of $\mathbb{R}^n$.

Exercise 4.24. Show that the linear span of any set of vectors in $\mathbb{R}^n$ is a linear subspace of $\mathbb{R}^n$.

Exercise 4.25. Let V be a linear subspace of $\mathbb{R}^n$ and S be a finite subset of V . Show that the linear span of S is contained in V , that is,

$$\text{span}(S) \subseteq V.$$

Exercise 4.26. Let $v_1, v_2, \dots, v_k \in \mathbb{R}^n$ be linearly independent vectors. Show that if $u \in \mathbb{R}^n$ is not in the span of $\{v_1, v_2, \dots, v_k\}$, then the set of vectors $\{v_1, v_2, \dots, v_k, u\}$ is linearly independent.

Exercise 4.27. Let $v_1, v_2, \dots, v_k$ be elements of $\mathbb{R}^n$. Show that if $u \in \mathbb{R}^n$ is in the span of $\{v_1, v_2, \dots, v_k\}$, then the set of vectors $\{v_1, v_2, \dots, v_k, u\}$ is linearly dependent.

Exercise 4.28. Let U, V be linear subspaces of $\mathbb{R}^n$. Show that the sum $U + V$ is equal to the sum $V + U$.

Exercise 4.29. Let U, V be linear subspaces of $\mathbb{R}^n$. Show that $U \subseteq U + V$ and $V \subseteq U + V$.

Exercise 4.30. Let V be a linear subspace of $\mathbb{R}^n$. Show that $V + V = V$.

Exercise 4.31. Let U, V be the subsets of $\mathbb{R}^3$ defined as

$$U = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + 2y = 0 \right\},$$

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}.$$

Determine the sum of U and V .

Exercise 4.32. Show that the sum of two linear subspaces of $\mathbb{R}^n$ is also a linear subspace of $\mathbb{R}^n$.

Exercise 4.33. Provide examples of linear subspaces V_1, V_2, V_3 of $\mathbb{R}^3$ satisfying

$$\{0\} \subsetneq V_1 \subsetneq V_2 \subsetneq V_3 \subseteq \mathbb{R}^3.$$

Exercise 4.34. Let e_1, e_2 denote the standard basis vectors of $\mathbb{R}^2$, that is,

$$e_1 := \begin{pmatrix} 1 \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Show that each of the following sets of vectors generates $\mathbb{R}^2$:

1. $\{e_1, e_2\}$,
2. $\{e_1, e_1 + e_2\}$,
3. $\{e_2, e_1 + e_2\}$,
4. $\{e_1 - e_2, e_1 + e_2\}$,
5. $\{e_1, e_2, e_1 + e_2\}$.

Prove that

$$\{ae_1 + be_2, ce_1 + de_2\}$$

generates $\mathbb{R}^2$ if and only if $ad - bc \neq 0$.

Exercise 4.35. Show that the linear subspace $\{0\}$ of $\mathbb{R}^n$ is generated by the set $\{0\}$.

Exercise 4.36. Determine a generating set for the subspace

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}.$$

Exercise 4.37. Determine a generating set for each of the subspaces as in the prior examples or exercises.

Exercise 4.38. Show that $\mathbb{R}^n$ is generated by the standard basis vectors $\{e_1, e_2, \dots, e_n\}$, where

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, e_n := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

The following exercise shows that

$$\text{span}(S) + \text{span}(T) = \text{span}(S \cup T)$$

for any finite sets of vectors S and T in $\mathbb{R}^n$.

Exercise 4.39. Let U, V be linear subspaces of $\mathbb{R}^n$. Suppose U is generated by the set of vectors $\{u_1, u_2, \dots, u_s\}$ and V is generated by the set of vectors $\{v_1, v_2, \dots, v_t\}$. Show that the sum $U + V$ is generated by the set of vectors $\{u_1, u_2, \dots, u_s\} \cup \{v_1, v_2, \dots, v_t\}$.

Exercise 4.40. Show that the standard basis vectors

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, e_n := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

form a basis of $\mathbb{R}^n$.

Exercise 4.41. Determine whether the subsets of $\mathbb{R}^2$ as in [Exercise 4.34](#) are bases of $\mathbb{R}^2$.

Exercise 4.42. Determine a basis of the linear subspace

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x - 2y + 6z = 0 \right\}.$$

of $\mathbb{R}^3$.

Exercise 4.43. Determine a basis of the linear subspace

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y = 0 \right\}.$$

of $\mathbb{R}^3$.

Exercise 4.44. Determine a basis of each of the subspaces as in the prior examples or exercises.

Exercise 4.45. Show that $\dim(\mathbb{R}^n) = n$.

Exercise 4.46. Determine a basis and the dimension of each of the subspaces as in the prior examples or exercises.

Practice Problems

Chapter 5. Sets

Exercise 5.1. Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$. Find

$$A \cap B, A \cup B, A \setminus B, B \setminus A, A \Delta B.$$

Exercise 5.2. Let $U = \{1, 2, \dots, 10\}$ be the universal set. If $A = \{2, 4, 6, 8, 10\}$ and $B = \{1, 3, 5, 7, 9\}$, then find

$$A^c, B^c, (A \cup B)^c, (A \cap B)^c.$$

Exercise 5.3. Write down the power set of each of the following sets:

$$\{1, 2\}, \{1, 2, 3\}, \{1, 2, 3, 4\}.$$

Exercise 5.4. Identify the set

$$\left\{x \in \mathbb{R} \setminus \{0\} : x + \frac{1}{x} \geq 2\right\}.$$

Exercise 5.5. Let A, B be subsets of $\mathbb{R}$ defined by

$$A = \{x \in \mathbb{R} : x^2 \geq 0\},$$

$$B = \{x \in \mathbb{R} : x^3 \geq 0\}.$$

Determine the sets $A \cup B, A \cap B, A \setminus B, B \setminus A$.

Exercise 5.6. For a positive integer k , let A_k denote the set of integral multiples of k , that is,

$$A_k := \{r \in \mathbb{Z} : r = k\ell \text{ for some } \ell \in \mathbb{Z}\}.$$

Let m, n be positive integers. For each of the following statements, determine the equivalent conditions on the integers m, n .

1. $A_m \subseteq A_n$
2. $A_m \subsetneq A_n$
3. $A_m \not\subseteq A_n$
4. $A_m = A_n$

Chapter 6. Induction

Exercise 6.1. Show that

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

for all $n \in \mathbb{N}$.

Exercise 6.2. Show that

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

for all $n \in \mathbb{N}$.

Exercise 6.3. Show that

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \cdots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$$

for all $n \in \mathbb{N}$.

Exercise 6.4. Show that

$$3 + 11 + \cdots + (8n - 5) = 4n^2 - n$$

for all $n \in \mathbb{N}$.

Exercise 6.5. Show that

$$1^2 + 3^2 + \cdots + (2n-1)^2 = \frac{4n^3 - n}{3}$$

for all $n \in \mathbb{N}$.

Exercise 6.6. Show that

$$1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{n+1}n^2 = (-1)^{n+1} \frac{n(n+1)}{2}$$

for all $n \in \mathbb{N}$.

Exercise 6.7. Show that

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6},$$

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

hold for any positive integer n .

Exercise 6.8. Show that

$$\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2n-1} - \frac{1}{2n}$$

for all $n \in \mathbb{N}$.

Exercise 6.9. Show that $n^3 + 5n$ is divisible by 6 for all $n \in \mathbb{N}$.

Exercise 6.10. Show that $5^{2n} - 1$ is divisible by 8 for all $n \in \mathbb{N}$.

Exercise 6.11. Show that $5^n - 4n - 1$ is divisible by 16 for all $n \in \mathbb{N}$.

Exercise 6.12. Show that $6^n - 5n - 1$ is divisible by 25 for all $n \in \mathbb{N}$.

Exercise 6.13. Show that $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9 for all $n \in \mathbb{N}$.

Exercise 6.14. Show that

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

for all $n \in \mathbb{N}$ satisfying $n > 1$.

Chapter 7. Matrices

Exercise 7.1. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 1 \\ 1 & 6 \end{pmatrix}$. Compute $2A + 3B$, A^2B , AB^3 , A^T .

Exercise 7.2. Suppose A is a 3×2 matrix, and

$$A \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 8 \\ 14 \end{pmatrix},$$

$$A \begin{pmatrix} 3 \\ 7 \end{pmatrix} = \begin{pmatrix} -4 \\ 8 \\ 20 \end{pmatrix}.$$

Determine

$$A \begin{pmatrix} 103 \\ 407 \end{pmatrix}, A \begin{pmatrix} 97 \\ 393 \end{pmatrix}.$$

Exercise 7.3. Show that every vector in $\mathbb{R}^2$ can be expressed as a linear combination of the vectors

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix}.$$

Exercise 7.4. Write the following system of equations in matrix form $Ax = b$.

$$2x + 3y = 5,$$

$$4x - y = 1.$$

Determine if A is invertible. If it is, find A^{-1} and use it to solve the system.

Exercise 7.5. If A, B are 2×2 matrices, then show that

$$\det(AB) = \det(A) \det(B).$$

Exercise 7.6. Determine which of the following matrices are upper-triangular, lower-triangular, or neither.

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}, \begin{pmatrix} 7 & 8 \\ 0 & 9 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{pmatrix}, \begin{pmatrix} 7 & 0 \\ 8 & 9 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \\ 5 & 0 & 6 \end{pmatrix}.$$

Exercise 7.7. Determine which of the following matrices are symmetric, which are skew-symmetric, and which are neither.

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}, \begin{pmatrix} 7 & 8 \\ 8 & 9 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix},$$

$$\begin{pmatrix} 0 & 2 & 3 \\ -2 & 0 & 5 \\ -3 & -5 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 8 \\ -8 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Exercise 7.8. Determine which of the following matrices are orthogonal.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Exercise 7.9. Let

$$A = \begin{pmatrix} 1 & -5 & 6 \\ -4 & 8 & -9 \\ 7 & 2 & 10 \end{pmatrix}.$$

Find $\det(A)$ and $\text{adj}(A)$.

Exercise 7.10. Let

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 3 & 2 & 1 \\ 4 & 1 & 2 \end{pmatrix}.$$

Find $\det(A)$ and $\text{adj}(A)$.

Exercise 7.11. Let θ be a real number satisfying $0 < \theta < \frac{\pi}{2}$. Show that the matrix

$$A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

is orthogonal.

Exercise 7.12. Let A be an orthogonal matrix. Show that A is invertible and that

$$A^{-1} = A^T.$$

Exercise 7.13. Write the following system of equations in matrix form $Ax = b$.

$$\begin{aligned} x_1 + 2x_2 - x_3 + 4x_4 &= 5, \\ 2x_1 + 3x_2 + x_3 + 5x_4 &= 8, \\ -x_1 + 4x_2 - 2x_3 - 3x_4 &= -4, \\ 3x_1 - x_2 + 4x_3 + 2x_4 &= 7. \end{aligned}$$

Perform Gaussian elimination on the augmented matrix $(A \mid b)$, to reduce it to row echelon form. Using this form, determine if the system has a solution. If it does, find all solutions of the above system in the variables x_1, x_2, x_3, x_4 .

Exercise 7.14. Perform row reduction on the matrix

$$\begin{pmatrix} 1 & 2 & -5 \\ 2 & 5 & -7 \\ -1 & -2 & 4 \end{pmatrix}$$

to reduce it to row echelon form. Using these reductions, find the inverse of the above matrix, if it exists.

Exercise 7.15. Transform the matrix

$$\begin{pmatrix} 0 & 2 & -3 & 0 & 0 \\ 1 & -1 & 4 & 0 & 5 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 3 & -5 & 0 & 1 \\ 0 & 3 & -5 & -1 & 2 \end{pmatrix}$$

into a matrix in row echelon form using elementary row operations.

Exercise 7.16. Solve the system of equations in the variables x_1, x_2, x_3, x_4 .

$$\begin{aligned} 2x_2 - 3x_3 &= 0, \\ x_1 - x_2 + 4x_3 &= 5, \\ x_4 &= -2, \\ 3x_2 - 5x_3 &= 1, \\ 3x_2 - 5x_3 - x_4 &= 2. \end{aligned}$$

Exercise 7.17. Let

$$A = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix},$$

where a, b, c are scalars. Show that $\det(A) = abc$, and that

$$\operatorname{adj}(A) = \begin{pmatrix} bc & 0 & 0 \\ 0 & ca & 0 \\ 0 & 0 & ab \end{pmatrix}.$$

Exercise 7.18. Compute the determinant of the matrix

$$A = \begin{pmatrix} 1 & -5 & 6 \\ 2 & 0 & 9 \\ -1 & 2 & -1 \end{pmatrix}$$

to determine whether it is invertible. If A is invertible, then find its inverse using its adjoint.

Exercise 7.19. Use Gaussian elimination to determine the row echelon form of the matrix

$$A = \begin{pmatrix} 1 & -5 & 6 \\ 2 & 0 & 9 \\ -1 & 2 & -1 \end{pmatrix}.$$

Use the row echelon form to determine whether A is invertible. If A is invertible, then find its inverse using elementary matrices, corresponding to the elementary row operations used to transform A into its row echelon form.

Exercise 7.20. Solve the following system of equations in the variables x, y, z .

$$\begin{aligned} x + 2y + 3z &= 1, \\ 4x - 5y + 6z &= 2, \\ 7x - y + 10z &= 3. \end{aligned}$$